This article was downloaded by: [129.186.138.35] On: 08 February 2019, At: 05:32
Publisher: Institute for Operations Research and the Management Sciences (INFORMS)
INFORMS is located in Maryland, USA

Manufacturing & Service Operations Management

Publication details, including instructions for authors and subscription information:
http://pubsonline.informs.org

Dynamic Pricing of Omnichannel Inventories

Pavithra Harsha, Shivaram Subramanian, Joline Uichanco

To cite this article:
Pavithra Harsha, Shivaram Subramanian, Joline Uichanco (2019) Dynamic Pricing of Omnichannel Inventories. Manufacturing & Service Operations Management

Published online in Articles in Advance 06 Feb 2019

. https://doi.org/10.1287/msom.2018.0737

Full terms and conditions of use: https://pubsonline.informs.org/page/terms-and-conditions

This article may be used only for the purposes of research, teaching, and/or private study. Commercial use or systematic downloading (by robots or other automatic processes) is prohibited without explicit Publisher approval, unless otherwise noted. For more information, contact permissions@informs.org.

The Publisher does not warrant or guarantee the article's accuracy, completeness, merchantability, fitness for a particular purpose, or non-infringement. Descriptions of, or references to, products or publications, or inclusion of an advertisement in this article, neither constitutes nor implies a guarantee, endorsement, or support of claims made of that product, publication, or service.

Copyright © 2019, INFORMS

Please scroll down for article—it is on subsequent pages

INFORMS is the largest professional society in the world for professionals in the fields of operations research, management science, and analytics.
For more information on INFORMS, its publications, membership, or meetings visit http://www.informs.org

http://pubsonline.informs.org/journal/msom/

MANUFACTURING & SERVICE OPERATIONS MANAGEMENT
Articles in Advance, pp. 1–19
ISSN 1523-4614 (print), ISSN 1526-5498 (online)

Honorable Mention—2017 *M&SOM* Practice-Based Research Competition

Dynamic Pricing of Omnichannel Inventories

Pavithra Harsha,[a] Shivaram Subramanian,[a] Joline Uichanco[b]

[a] IBM T. J. Watson Research Center, Yorktown Heights, New York 10598; [b] Ross School of Business, University of Michigan, Ann Arbor, Michigan 48109

Contact: pharsha@us.ibm.com, https://orcid.org/0000-0002-6049-7739 (PH); subshiva@us.ibm.com (SS); jolineu@umich.edu, https://orcid.org/0000-0003-2110-7088 (JU)

Received: September 14, 2016
Revised: April 28, 2017; January 10, 2018; March 21, 2018
Accepted: June 1, 2018
Published Online in Articles in Advance: February 6, 2019

https://doi.org/10.1287/msom.2018.0737

Copyright: © 2019 INFORMS

Abstract. Omnichannel retail refers to a seamless integration of an e-commerce channel and a network of brick-and-mortar stores. An example is cross-channel fulfillment, which allows a store to fulfill online orders in any location. Another is price transparency, which allows customers to compare the online price with store prices. This paper studies a new and widespread problem resulting from omnichannel retail: price optimization in the presence of cross-channel interactions in demand and supply, where cross-channel fulfillment is exogenous. We propose two pricing policies that are based on the idea of "partitions" to the store inventory that approximate how this shared resource will be utilized. These policies are practical because they rely on solving computationally tractable mixed integer programs that can accept various business and pricing rules. In extensive simulation experiments, they achieve a small optimality gap relative to theoretical upper bounds on the optimal expected profit. The good observed performance of our pricing policies results from managing substitutive channel demands in accordance with partitions that rebalance inventory in the network. A proprietary implementation of the analytics is commercially available as part of the IBM Commerce markdown price solution. The system results in an estimated 13.7% increase in clearance-period revenue based on causal model analysis of the data from a pilot implementation for clearance pricing at a large U.S. retailer.

Funding: J. Uichanco acknowledges partial support by the National Science Foundation–Civil, Mechanical and Manufacturing Innovation [Grant 1561791].
Supplemental Material: The e-companion is available at https://doi.org/10.1287/msom.2018.0737.

Keywords: omnichannel • pricing • attraction demand • markdown pricing • e-commerce fulfillment • cross-channel • elasticity

1. Introduction

Omnichannel retailing is a recent trend sweeping companies across the retailing industry. An omnichannel strategy promises to revolutionize how companies engage with customers by creating a seamless shopping experience through an alignment of the retailer's multiple sales channels. The following are a few examples of the new capabilities enabled by omnichannel retail:

- A customer can buy a product from the online store while the customer is in a brick-and-mortar store after finding through the customer's mobile phone that it is offered at a cheaper price online.
- A customer who purchased a product online might choose a "buy online, pick up in store" option to receive the product sooner than waiting for the package to be shipped to the customer's address.
- The package that an e-commerce customer receives might have been fulfilled from a nearby store because the e-commerce fulfillment center is out of stock.

From a customer's perspective, an omnichannel environment makes it easier to compare the store price to the online price, to purchase a product from any channel, and to receive the product through any of the retailer's channels. These conveniences allow the retailer to be competitive in a crowded e-commerce market. Additionally, with omnichannel integration, a retailer can use its existing store infrastructure to ship items to its online customers. Amazon.com, the largest e-commerce retailer in the United States, has a large number of fulfillment centers that are close to customer locations to allow fast and efficient fulfillment. In contrast, primarily brick-and-mortar retailers only have a few fulfillment centers because of the significant fixed cost. Thus, many omnichannel retailers rely primarily on cross-channel fulfillment through ship-from-store (SFS) fulfillment and buy online, pick up in store to increase the speed of delivery to its online customers.

Despite omnichannel integration providing many benefits, it also introduces new challenges for price optimization. Traditional retail pricing models optimize channel prices under the assumption that there is no inventory sharing and coordination between channels. This assumption does not hold in omnichannel with which store inventory is used for fulfilling customer orders placed online. Another challenge is the potential demand substitution between the online and brick-and-mortar stores, which is affected by the prices offered on

the two channels. Channel substitution is ignored in traditional revenue management models, which assume price only affects demand in the same channel. Despite these new challenges, many omnichannel retailers utilize legacy price optimization systems that do not account for any channel interdependencies.

We study a widespread and new problem resulting from omnichannel retail: price optimization on an omnichannel network in the presence of cross-channel interactions in demand and supply with which *cross-channel fulfillment is exogenous and unknown*. This is motivated from a business problem faced by a major U.S. omnichannel retailer, who we engaged with in a partnership with IBM Commerce, a merchandising solution provider. The partnership was formed to develop an omnichannel clearance-pricing optimization system to replace the retailer's legacy system that ignores cross-channel interactions. For this retailer, and for many other omnichannel retailers, it is difficult to fully coordinate pricing with fulfillment because order fulfillment is managed by a separate and independent operational solution called the order management system (OMS) that fulfills e-commerce orders on the fly, and price changes have a slower cadence (e.g., weekly). This problem is different from other models in revenue management literature in which prices have to be optimized, either for a dedicated resource or for an endogenously allocated shared resource.

We propose two pricing policies that are based on the idea of "partitions" to the store inventory that approximate how this shared resource will be utilized:

- Policy D-OCPX determines inventory partitions by optimizing the profit of a deterministic model in which all random variables are replaced with their expected values.
- Policy R-OCPX determines inventory partitions by optimizing the worst-case realized profit in an uncertainty set of the demand realizations.

Here OCPX refers to omnichannel pricing (OCP) with cross-channel (X) interactions. We demonstrate that these pricing policies are practical because they rely on solving computationally tractable mixed-integer programs (MIPs) that can be solved by commercial off-the-shelf optimization solvers. In experiments on data from the retailer, the omnichannel price-optimization model solves within 40 seconds for most problem instances, which is well within the business cycle requirement for markdown optimization (MDO). Various business rules and pricing rules of a retailer can be included as constraints to these models. Importantly, the policies are shown to perform well in extensive simulation experiments and in a pilot implementation of the omnichannel markdown-optimization system with the partner retailer. A rigorous empirical analysis of data resulting from the pilot implementation of the omnichannel system shows an estimated 13.7% increase in clearance period revenues resulting from the new omnichannel system. The good observed performance of our pricing policies results from managing demand substitution across channels in accordance with partitions that rebalance inventory in the network. In particular, compared with the legacy policy, our proposed policy results in higher online demand, which can be used to rebalance inventory by choosing stores with excess inventory for online fulfillment.

Because of the positive feedback from the retailer during the engagement and the widespread nature of the problem, a proprietary implementation of the proposed analytics was commercialized in May 2016 and is now available as part of the IBM Commerce markdown price solution.

1.1. Literature Review

Omnichannel retailing is a relatively new area; hence, there are few academic papers on optimization of omnichannel operations. There are past studies on multichannel pricing in marketing literature, and surveys by Zhang et al. (2010) and Grewal et al. (2011) provide a good overview. Our work differs in that our goal is to develop a computationally tractable model and a decision-support system for omnichannel pricing. Broadly, our work is in the area of revenue management, in which most previous works either adopt price control (use prices to control demand) or capacity control (allocate capacity of resources to classes of demand). A novel aspect of our work is that it employs both: cross-channel demand substitution that is managed through pricing and allocation (partition) of capacities optimally across channels. In addition, the capacities themselves are substitutable as online demand can be fulfilled using inventory from any store.

For comprehensive surveys of dynamic pricing, see Bitran and Caldentey (2003), Elmaghraby and Keskinocak (2003), and Chen and Simchi-Levi (2012). Classical dynamic-pricing models consider a finite amount of perishable inventory being sold to price-sensitive customers over a finite horizon. If demand follows a Poisson process with known price-dependent intensity, Gallego and van Ryzin (1994), Bitran and Mondschein (1997), and Zhao and Zheng (2000) characterize the optimal pricing policy. Caro and Gallien (2012) discuss the development and implementation of a clearance-pricing model for the fast-fashion retailer Zara. Bitran et al. (1998) extend the basic dynamic-pricing model to a network of brick-and-mortar stores that coordinate prices and trans-ship inventory between stores. Although the model allows for inventory flows between locations (similar to cross-channel fulfillment), it considers a single channel and no demand substitution. A paper by Harsha et al. (2019) studies price optimization for omnichannel demand with channel substitution, but their work ignores inventory constraints or cross-channel fulfillment flows. Another related topic is dynamic pricing of multiple products that share resources, and each product consumes a fixed

capacity of the resources (Gallego and van Ryzin 1997, Maglaras and Meissner 2006). In omnichannel, the channel location is the "product" and the inventories in different locations are "resources," but consumption is not known a priori because it is determined by cross-channel fulfillment decisions.

Most of these referenced papers assume that the demands can be characterized completely using a probability-distribution function. But some recent papers have also considered formulations and models that avoid specifying complete distributional information for the unknown parameters. One such approach is using robust optimization formulations based on either maximizing the minimum possible revenue (Adida and Perakis 2006, Thiele 2009), minimizing the worst-case regret (Perakis and Roels 2008), or maximizing the competitive ratio (Lan et al. 2008).

A related stream of literature is the topic of network-capacity control wherein, unlike price-based controls, the goal is to allocate resources across products at fixed prices by deciding which subset of products to offer. Store-inventory partitions introduced in our paper are similar in spirit to booking limits or protection limits in capacity-control problems. Early papers in this topic assume product independence (Talluri and van Ryzin 1999, Topaloglu 2009), and consumer substitution across products is more recently studied (Liu and van Ryzin 2008, Bront et al. 2009). Cross-channel fulfillment is related to supply-side substitution in capacity-control literature. Gallego and Phillips (2004) introduce the idea of a "flexible product," which is a menu of alternative products. The customer purchases a flexible product knowing that the seller will assign the final product later. In omnichannel fulfillment, products purchased online are flexible products because the retailer can later fulfill the purchase from any store location.

We mention a few papers on pricing and/or fulfillment for a pure e-commerce retailer. Acimovic and Graves (2014) discuss the development of a dynamic-fulfillment system to optimize outbound shipping costs (affected by shipping distances and split shipments). Jasin and Sinha (2015) propose heuristics based on solving a deterministic linear-program (DLP) approximation. A paper by Lei et al. (2017) studied dynamic joint pricing and fulfillment decisions of a pure e-commerce retailer.

2. The Omnichannel Markdown Optimization Problem

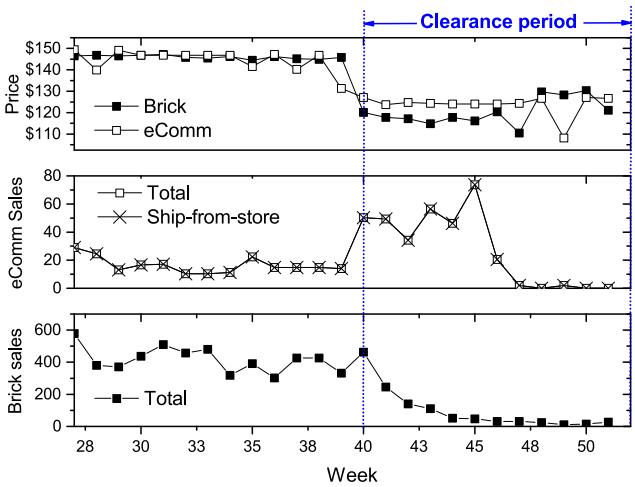
The partner retailer generates more than $40 billion in sales annually through an online store and its more than 1,000 brick-and-mortar stores in the United States as of 2014. Although clearance sales is a small fraction of the retailer's total sales, it is significant in absolute terms (more than $1 billion per year). Table 1 summarizes data for a sample of 195 clearance stock keeping units (SKUs) in 2014 from the partner retailer. About 89% of sales made in these categories is through store purchase, and about 11% is through the online channel. Almost 94% of online sales are fulfilled using inventory shipped from a store. During clearance alone, the online sales share steeply increases to about 24% and the SFS fraction is close to 100%.

With omnichannel operations evident from Table 1, using legacy markdown-optimization systems that ignored cross-channel interactions results in negative consequences. Figure 1 shows weekly channel prices (e-commerce price and average brick-and-mortar price) and weekly channel sales for a tablet computer that has been marked for clearance in both channels starting from week 40. At the start of the clearance period, the legacy pricing software sets large initial markdowns in the brick-and-mortar stores to clear off all store inventory over the next 12 weeks from walk-in store customers (see topmost panel). However, not all store inventory is at risk of becoming unsold because e-commerce sales are primarily fulfilled by stores during the clearance period (see middle panel). Because of the large store markdowns, after an initial sales spike on week 40, total brick-and-mortar sales steadily decline because of an increasing number of stores stocking out (see bottom panel). In contrast to the large brick-and-mortar markdowns, the legacy system initially sets no markdown for the online channel because inventory in the e-commerce warehouse is depleted by week 40; hence, the system assumes there is no inventory left for online sales. However, because of substitution, setting the online price significantly higher than the brick-and-mortar price will cause channel cannibalization. Merchandise managers would override the price outputs from the system to reduce this price difference by making ad hoc adjustments to the system inputs. A common

Table 1. Data Summary of a Sample of 195 Clearance SKUs in 2014 from Partner Retailer

Categories	# SKUs	Clearance sales, million units	Clearance revenue, $ millions	Percentage sales from e-commerce	Percentage e-commerce ship from store
Notebooks	100	4.1	84	13.4	94.6
Tablets	45	1.4	17	14.4	93.7
Tablet accessories	50	2	6	5.4	90.4
Total	195	7.5	107	11.4	93.8

Figure 1. (Color online) Time Series Data for the Channel Prices and the Channel Sales of a Tablet SKU



adjustment used to set the online price (also applied to the tablet example) is to inject artificial inventory into the online channel. Merchandise managers we interviewed described this manual adjustment process as unmanageable, labor intensive, and time consuming. Additionally, a manual adjustment process cannot be used for each of the more than 1,000 stores. Hence, a major concern was that, because most sales occurred through stores, the large brick-and-mortar markdowns would result in system-wide margin erosion.

2.1. Stochastic Optimization Model

We next model the omnichannel retailer's problem through a dynamic stochastic optimization model. We use the notation $[N]$ to denote the set $\{1, 2, \ldots, N\}$ for any integer $N \geq 1$. Any variables with tilde notation denote random variables. Subscripts e and b denote e-commerce and brick-and-mortar store variables, respectively.

Consider a retailer selling a product with finite inventory over T sales periods, and the length of a period is typically one week. Customers can purchase the product either through the retailer's online store or through any brick-and-mortar store in its network of retail stores. The retailer has one e-fulfillment center (EFC) that can fulfill any online purchase. Suppose that the retailer has stores in Z geographical zones, and we assume there is one retail store for each zone. The proposed model can be trivially extended to the case with multiple EFC locations, multiple stores per zone, or no stores in some zones. We assume that customers in region $z \in [Z]$ can only purchase from the online store or from the store in region z and are not willing to travel to a store in another region $z' \neq z$. We assume a short selling period; hence, inventory is not replenished in the EFC or in the stores. This assumption is suitable for seasonal, fast-fashion items or for clearance products. Because of the short horizon, we also assume that there are no inventory-holding costs and shortage costs. Inventory that is not sold by period T can be salvaged at a per-unit value q.

Because stores change prices infrequently, we assume the retailer can only change prices at the start of each week. At the start of week $t \in [T]$, let x_e^t be the inventory level in the EFC, and let x_{bz}^t be the inventory available in store $z \in [Z]$. After observing the period t inventory levels in the retail network $x^t = (x_e^t, x_{b1}^t, x_{b2}^t, \ldots, x_{bZ}^t)$, the retailer then determines the network-wide prices. A business rule of the partner retailer is that the online price must be uniform across customer locations, but each individual brick-and-mortar store can set its own price. We denote the vector of network prices as $p^t = (p_e^t, p_{b1}^t, p_{b2}^t \ldots, p_{bZ}^t)$, where p_e^t is the online price and p_{bz}^t is the store price in zone $z \in [Z]$. We let Ω be a discrete set of all feasible prices. For example, retailers often restrict markdowns to fixed discount levels (e.g., 20%, 30%, 40%) or prices with magic number endings (e.g., those ending with \$0.99) to exploit customer psychology.

After setting the price, the random channel demands and fulfillments are realized. We denote by $\tilde{D}^t = (\tilde{D}_{ez}^t, \tilde{D}_{bz}^t)_{z \in [Z]}$ the random vector of demands, where $\tilde{D}_{mz}^t = \tilde{D}_{mz}^t(p_e^t, p_{bz}^t)$ is the random channel m demand in zone z, which depends on the online price and the store price in zone z. Note that this demand model allows substitution between the online and store channels. We denote $d_{mz}^t(p_e^t, p_{bz}^t) := E[\tilde{D}_{mz}^t(p_e^t, p_{bz}^t)]$ as its expectation. Let $\tilde{y}_{ez}^t$ be a random variable denoting the portion of zone z online demand that has been fulfilled by EFC inventory and $\tilde{y}_{z'z}^t$ be the random portion fulfilled by store inventory from zone $z' \in [Z]$; any unfulfilled portion is lost. Zone z' can either be the same zone as z or a different zone. Suppose that zone z online orders fulfilled from the EFC incur a per-unit fulfillment cost c_{ez}, and those fulfilled from a store in zone z' incurs a per-unit fulfillment cost $c_{z'z}$. The fulfillment costs include shipping and handling costs. Define $c = (c_{ez}, c_{1z}, c_{2z}, \ldots, c_{Zz})_{z \in [Z]}$. The fulfillment vector $\tilde{y}^t = (\tilde{y}_{ez}^t, \tilde{y}_{1z}^t, \tilde{y}_{2z}^t, \ldots, \tilde{y}_{Zz}^t)_{z \in [Z]}$ is a random vector because it is realized exogenously (to the model) by the retailer's OMS and cannot be jointly optimized with the prices.

Given the week t realization of demand $\tilde{D}^t$ and fulfillment $\tilde{y}^t$, the retailer realizes its period t profit through recourse sales variables. The channel sales in zone $z \in [Z]$ are denoted by s_{ez}^t and s_{bz}^t. We denote the vector of sales variables as $s^t = (s_{ez}^t, s_{bz}^t)_{z \in Z}$. The state variables for $t+1$ then become $\tilde{x}_e^{t+1} = x_e^t - \sum_{z \in [Z]} \tilde{y}_{ez}^t$ and $\tilde{x}_{bz}^{t+1} = x_{bz}^t - s_{bz}^t - \sum_{z' \in [Z]} \tilde{y}_{zz'}^t$ for all z. Hence, $\tilde{x}^{t+1} = f(x^t, s^t, \tilde{y}^t)$, where f is a linear mapping.

The optimal policy maximizes the total expected profit, which is the expected revenue minus the expected

fulfillment cost over the selling horizon of length T. It can be found by solving the following Bellman equations:

$$V_*^t(x^t) = \max_{p^t \in \Omega^{Z+1}} E_{\tilde{D}^t, \tilde{y}^t}\left(\max_{s^t \in \mathcal{S}^t(x^t, \tilde{D}^t, \tilde{y}^t)} p^{t\top} s^t - c^\top \tilde{y}^t + V_*^{t+1}\left(f(x^t, s^t, \tilde{y}^t)\right)\right), \quad t \in [T], \tag{1}$$

$$V_*^{T+1}(x^{T+1}) = q(e^\top x^{T+1}), \tag{2}$$

where V_*^t is the period t value function, e is the vector of all ones, and $\mathcal{S}^t$ is the set of all feasible sales variables given the period t inventory, demand, and fulfillment. However, there are practical challenges for solving the Bellman equations. First, the dynamic-programming equations suffer from the well-known curse of dimensionality. Second, it is difficult to determine a probability distribution for the random vector $\tilde{y}^t$ because it depends on a separate and independent order-management system that determines on the fly the fulfilling store or EFC primarily based on operational costs but may also include complex rules that depend on store performance, store traffic, store fulfillment capacity, minimum store inventory levels, local weather, and order-splitting rules.

3. Omnichannel Pricing Policies

We develop omnichannel pricing policies that overcome the challenges of solving the omnichannel markdown-optimization problem in which pricing and fulfillment cannot be fully coordinated. These policies use the idea of *inventory partition* variables to ration the amount of store inventory available for online fulfillment, similar to booking limits in capacity control. Unlike booking limits, however, the partitions cannot be enforced by accepting or rejecting fulfillment requests to a store. Rather, the policies utilize channel prices to adjust demand to meet the inventory partitions.

The legacy pricing system outputs suboptimally large store markdowns because of its assumption that store inventory is only used for store sales even though in reality it is also used for online fulfillment. Optimizing price markdowns on products used in omnichannel fulfillment require identifying partitions of store inventory units by their usage (e.g., store sales or online fulfillment). However, the true partitions are unknown because they depend on the exogenously determined fulfillment. Directly forecasting such partitions using historical data yields a 70%–80% mean absolute percentage error. To develop more accurate inventory partitions that would be used for the pricing policies, we introduce endogenous fulfillment variables y^t to approximate the exogenous fulfillment $\tilde{y}^t$ (see Section 4.3 for the resultant lower errors of this approach observed in the pilot). Particularly, we consider a model in which pricing and fulfillment are coordinated, resulting in the Bellman equations

$$V^t(x^t) = \max_{p^t \in \Omega^{Z+1}} E_{\tilde{D}^t}\left(\max_{(s^t, y^t) \in \mathcal{X}^t(x^t, p^t, \tilde{D}^t)} p^{t\top} s^t - c^\top y^t + V^{t+1}(f(x^t, s^t, y^t))\right), \quad t \in [T], \tag{3}$$

$$V^{T+1}(x^{T+1}) = q(e^\top x^{T+1}), \tag{4}$$

where $\mathcal{X}^t(x^t, p^t, \tilde{D}^t)$ is the feasible set of period t fulfillment variables and sales variables:

$$\mathcal{X}^t(x^t, p^t, \tilde{D}^t) = \left\{ s \geq 0, y \geq 0 \;\middle|\; \begin{array}{ll} s_{mz}^t \leq \tilde{D}_{mz}^t(p_e^t, p_{bz}^t), & m = e, b, \forall z \in [Z] \\ s_{ez}^t = y_{ez}^t + \sum_{z' \in [Z]} y_{z'z}^t, & \forall z \in [Z], \\ \sum_{z \in [Z]} y_{ez}^t \leq x_e^t, & \\ s_{bz}^t + \sum_{z' \in [Z]} y_{zz'}^t \leq x_{bz}^t, & \forall z \in [Z], \end{array} \right\}. \tag{5}$$

The first set of constraints in Equation (5) ensures that the sales variables do not exceed the realized demand. The second set ensures that all online sales are fulfilled. The remaining constraints ensure inventory levels in all periods are nonnegative. If $\tilde{D}_{mz}^t$ is a nonnegative random variable, then starting from nonnegative inventory levels x^1 at $t = 1$, the sequence of feasible sets $\mathcal{X}^t(\tilde{x}^t, \tilde{p}^t, \tilde{D}^t)$ for $t \in [T]$ are nonempty with probability one.

If the OMS fulfillment rules satisfy all constraints of $\mathcal{X}^t$, then $V^1(x^t)$ is an upper bound to the optimal expected omnichannel profit $V_*^1(x^t)$ because jointly optimizing prices and fulfillment achieves a higher expected profit. Because V_*^1 cannot be evaluated because of the unknown fulfillment model and V^1 cannot be solved because of the curse of dimensionality, we use upper bounds to V^1 as benchmarks to bound the optimality gap of omnichannel pricing policies. For example, in the following lemma, an upper bound to V^1 is a linear program (LP) if demand results from a Bernoulli arrival process, in which arrivals are split into the different channel zones. (In this case, the channel-zone demand in a given week has a multinomial distribution.)

Lemma 1. *Suppose period $t \in [T]$ channel-zone demand has a multinomial distribution with N^t as the number of trials and $\psi_{mz}^t(p_e, p_{bz})$ as the probability of a demand occurring in zone $z \in [Z]$ and channel $m \in \{e, b\}$ given the online price p_e and store price p_{bz}. Suppose that $\psi_{ez}^t(\infty, p) = \psi_{bz}^t(p, \infty) = 0$ for any p. Then $V^1(x) \leq V_{LP}$, where V_{LP} is the optimal value of a linear program.*

The proof of the lemma is in the e-companion. The idea behind the proof is that $V^1(x)$ is bounded by the optimal expected profit of a firm that can jointly coordinate

pricing and fulfillment with a continuous-time dynamic policy. The optimal profit in the latter setting is bounded by the optimal value of the deterministic linear program with the proof following similar lines as lemma 1 of Lei et al. (2017) but with nonstationary demand rates. A popular approach in revenue management under a continuous-time setting is the use of the optimal solution of the deterministic linear program to randomize the decision for each customer (Gallego and van Ryzin 1997, Liu and van Ryzin 2008, Jasin and Kumar 2013). For instance, Lei et al. (2017) use this approach to develop a randomized joint pricing and fulfillment policy for an e-commerce retailer, for which the profit converges to the optimal profit when both demand and capacity scale up. We find computational experiments later that show an adaptation of this randomized policy to the omnichannel setting, in which price can only be changed weekly and fulfillment is exogenous, is suboptimal. Hence, in what follows, we develop a new heuristic that performs well in the omnichannel setting.

3.1. Deterministic Omnichannel Pricing Policy (D-OCPX)

We approximate the value function V^t with the optimal value of a computationally tractable optimization model. Because the optimal inventory partitions not only depend on the fulfillment costs and salvage cost, but also on the future prices in the whole retail network, we approximate V^t by the optimal value V_D^t of a deterministic multiperiod optimization model that jointly optimizes prices $(p^t, p^{t+1}, \ldots, p^T)$ and optimal inventory partitions. In particular,

$$V_D^t(x^t) := \underset{p,s,y,u}{\text{maximize}} \quad \sum_{k=t}^{T} \sum_{z\in[Z]} (p_e^k s_{ez}^k + p_{bz}^k s_{bz}^k) - \sum_{z\in[Z]} c_{ez} y_{ez} - \sum_{z\in[Z]} \sum_{z'\in[Z]} c_{zz'} y_{zz'} + q\left(u_e + \sum_{z\in[Z]} u_{bz}\right) \tag{6a}$$

$$\text{subject to} \quad s_{mz}^k \le d_{mz}^k(p_e^k, p_{bz}^k), \quad k = t, \ldots, T, \quad m = e, b, \ \forall z \in [Z] \tag{6b}$$

$$\sum_{z\in[Z]} y_{ez} + u_e = x_e^t \tag{6c}$$

$$\sum_{k=t}^{T} s_{bz}^k + \sum_{z'\in[Z]} y_{zz'} + u_{bz} = x_{bz}^t, \quad \forall z \in [Z] \tag{6d}$$

$$\sum_{k=t}^{T} s_{ez}^k = y_{ez} + \sum_{z'\in[Z]} y_{z'z}, \quad \forall z \in [Z] \tag{6e}$$

$$s \ge 0, \ y \ge 0, \ u \ge 0 \tag{6f}$$

$$p^k \in \Omega^{Z+1} \quad k = t, \ldots, T. \tag{6g}$$

Variables y_{ez} and $y_{zz'}$ are the inventory partitions that determine the channel-zone use of EFC inventory and store inventory. Note that replacing all random demands in Equation (3) with their expected values and by setting $y_{ez} = \sum_{k=t}^{T} y_{ez}^k$ and $y_{zz'} = \sum_{k=t}^{T} y_{zz'}^k$ results in Equation (6). Hence, V_D^t is the "certainty equivalence" formulation of Equation (3).

Algorithm 1 is a pricing policy, which we refer to as D-OCPX based on these deterministic store inventory partitions. In each period, the pricing policy solves the optimization model (6) and sets the price according to its optimal solution. The optimality of D-OCPX under exogenous fulfillment depends on how accurately the fulfillment variables y^t approximate the actual fulfillment vectors $\tilde{y}^t$ from the OMS. In Section 4.3, we report the accuracy of this prediction in the commercial pilot implementation.

Algorithm 1. (Pricing Policy D-OCPX)

Require: A vector of initial inventory levels x^1 and feasible price set Ω

Ensure: Determine a sequence of online prices and zone-level store prices

(1) $t \leftarrow 1$
(2) **while** $t \le T$, **do**
(3) Solve (6) for the optimal solution (p^*, s^*, y^*, u^*)
(4) $p_e^t \leftarrow p_e^{t*}$
(5) $p_{bz}^t \leftarrow p_{bz}^{t*}$ for $z \in [Z]$
(6) Exogenously realize demands $D^t(p^t, \xi^t)$, sales s^t, and fulfillments y^t
(7) $x^{t+1} \leftarrow f(x^t, y^t, s^t)$
(8) $t \leftarrow t + 1$

We next discuss the computational tractability of D-OCPX. In each period t, the algorithm solves model (6), which is nonconvex because the objective function is bilinear and the feasible set could be nonconvex because of constraint (6b). However, because the feasible price set is discrete, the next lemma shows that V_D^t can be reformulated as a tractable MIP. Furthermore, the size of the MIP can be reduced under a multinomial logit (MNL) demand model commonly used in revenue management literature to model substitution among choices.

Lemma 2. *$V_D^t(x^t)$ is equal to the optimal value of a mixed-integer program with $O(I^2)$ binary variables. Under a multinomial logit demand model, the number of binary variables is $O(I)$.*

In the MNL demand model, given an arrival in zone z, if the price online and in store are p_e and p_b, respectively,

the probability that the customer will purchase from channel m is

$$\theta_{mz}(p_e, p_b) = \frac{\exp(\alpha_{mz} - \beta_{mz} p_m)}{1 + \exp(\alpha_{ez} - \beta_{ez} p_e) + \exp(\alpha_{bz} - \beta_{bz} p_b)}, \quad \text{for } m = e, b, \tag{7}$$

and the expected demand is $N_z \theta_{mz}(p_e, p_b)$. Here, N_z is the expected number of arrivals to zone z, and α_{ez}, $\alpha_{bz}, \beta_{ez}, \beta_{bz}$ are the choice probability parameters, all of which can be time period k-dependent. In our computational experiments later in Section 4 when implementing this model in CPLEX for realistic problem sizes, the solver with its default termination criteria solves most instances in less than 40 seconds.

We next demonstrate the performance of D-OCPX in simulations under a multinomial distribution of weekly demand. We choose $Z = 5$ and $T = 5$. At the beginning of week $t \in [T]$, the simulation sets the prices $p^t = (p_e^t, p_{b1}^t, \ldots, p_{b5}^t)$ based on the solution of Equation (6). Week t demand results from the following process. A total of N^t customers arrive, and an arrival occurs in zone z with probability N_z^t / N^t. Further, a zone z arrival chooses to purchase from a channel according to MNL probabilities. Hence, the weekly demand follows a multinomial distribution with N^t trials and success probabilities $\psi_{mz}^t(p_e, p_b) = N_z^t / N^t \cdot \theta_{mz}(p_e, p_b)$ for $z \in [Z]$; $m = e, b$ and choice probabilities $\theta_{mz}(p_e, p_b)$ are given in Equation (7). We use the following fulfillment rule for each incoming customer demand: if the customer chooses to purchase from the store, the sale is fulfilled using store z inventory if available. Otherwise, the sale is lost. If the customer chooses to purchase online, the sale is fulfilled by the store with the lowest fulfillment cost out of all stores with positive inventory.

We generated parameters for Equation (7) and N_z^t according to the method we describe later in Section 5.1, then set $N^t = \sum_{z \in [Z]} N_z^t$. The baseline total arrivals $N := \sum_{t \in [T]} N^t$ are 50. We set $x_e = 0$ and, for all $z \in [Z]$, set $x_{bz} = 0.5 \times \sum_{t \in [T]} N_z^t$ so that the total inventory is less than the expected number of arrivals in the zone. In different runs of the experiment, we report the performance of D-OCPX as the problem size increases by scaling the number of arrivals and the total inventory to σN and σx, respectively, where $\sigma \in \{1, 2, 4, 8, 16, 20, 24, 34\}$ is a scaling parameter. The feasible price set is between \$125 and \$500 (with a step size of \$12.50), and the salvage value is \$50. The distance between zones is randomly chosen, which determines the fulfillment costs. To test the pricing policies, we chose high fulfillment costs (\$46.70 for in zone, \$50.70–\$71.20 for cross-zone).

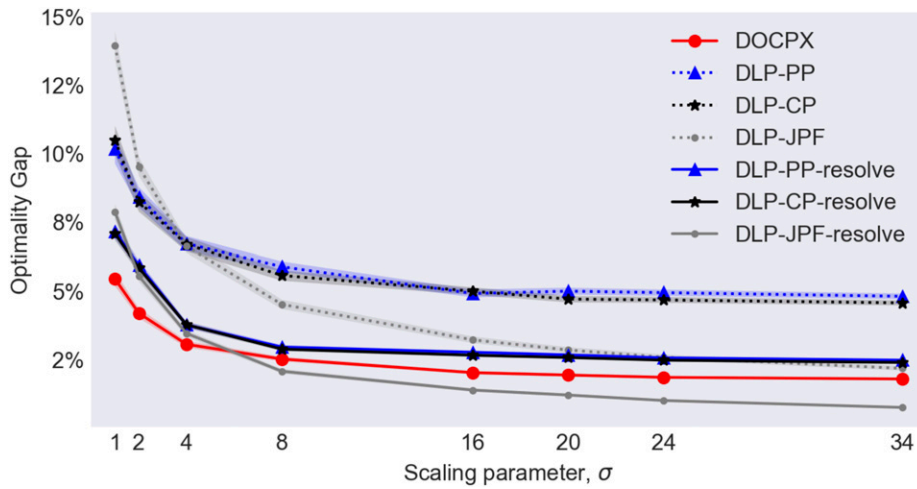
For each σ, we estimate the expected profit by averaging the profit of D-OCPX on 10^3 randomly generated sample paths, in which each sample path is a different sequence of the N arrivals. The optimality gap of D-OCPX is the relative difference of its sample average profit to the deterministic LP upper bound (see Lemma 1). We also compare D-OCPX to the following policies:

- Policy DLP-PP has periodic price changes at the start of each week t. It solves the deterministic LP on week 1 only. It uses the fractional solution to construct a probability distribution of the online price used to choose a random week t online price, then constructs the probability distribution for store prices (conditional on the online price) used to draw the week t store prices.
- Policy DLP-PP-resolve is the same as DLP-PP but with the LP model re-solved at the start of each week.
- Policy DLP-CP has continuous price changes (for each new customer). It solves the deterministic LP on week 1 only. It uses the LP solution to construct the joint probability distribution of the online price and store prices, from which it draws a *new* random online price and store price for *each* arriving customer.
- Policy DLP-CP-resolve is the same as DLP-CP but with the LP model re-solved at the start of each week.

Figure 2 reports the optimality gap (with 5% and 95% confidence levels) of the pricing policies plotted against the problem size-scale parameter σ. To verify our experiments, we also report the optimality gap of the joint pricing and fulfillment DLP continuous-time policy (DLP-JPF and DLP-JPF-resolve), which is the setting studied in Lei et al. (2017). Because this setting was used to compute the LP upper bound in Lemma 1, the bound applies for these policies. Observe from Figure 2 that the optimality gap of the DLP policies improves when the deterministic LP model is solved periodically. Hence, we focus our next discussion on the policies with re-solving.

A property of the continuous-time joint pricing and fulfillment DLP policy is its asymptotic optimality to the stochastic problem as demand and capacity are scaled up (Lei et al. 2017). The deterministic LP optimal value is a good approximation to this randomized policy's expected profit because expected demand approximates expected sales (demand can be turned off by choosing an infinite price whenever inventory runs out). The approximation is better for large sales volumes and long time periods in which fluctuations in sales over the season are "smoothed out." However, when price can only be changed periodically, as in the omnichannel setting, there is no guarantee that the expected demand in each period is close to the expected sales. Hence, the LP-optimal value does not approximate the expected profit of a randomized policy, and DLP-PP performs poorly. Interestingly, a DLP policy

Figure 2. (Color online) Optimality Gap of D-OCPX, DLP-PP, DLP-PP-resolve, DLP-CP, and DLP-CP-resolve Against the Problem Scale



Notes. DLP-JPF is also plotted as a reference. The optimality gap is the relative difference between the expected profit and V_{LP}.

with continuous price changes (DLP-CP) also performs poorly when fulfillment is exogenous. This is also because of expected demand not matching the expected sales, which is highly dependent on fulfillment. In fact, D-OCPX outperforms both policies under exogenous fulfillment.

3.2. Robust Omnichannel Pricing (R-OCPX)

Policy D-OCPX approximates the value function by assuming the stochastic demand would equal its expected value. Given the omnichannel prices, the inventory partition variables y_{ez} and $y_{zz'}$ in Equation (6) are aggregate fulfillment quantities that minimize the fulfillment cost of the *deterministic* demand. In the experiments with multinomial demand, D-OCPX has a low optimality gap. But in general, D-OCPX could have a higher optimality gap under highly variable demand distributions because the partition variables may be poor approximations of the optimal fulfillment quantities. In this case, ignoring uncertainty results in fulfillment decisions that incur significant lost in-store sales and high fulfillment costs (if demand is higher than expected) or earn low revenue (if demand is lower than expected). We discuss how to enhance pricing policy D-OCPX with inventory partitions that incorporate uncertainty through a robust optimization approach.

The robust approach assumes that, after the firm chooses the omnichannel prices, an "adversary" chooses demand realizations from an *uncertainty set* that result in the worst-case firm profit. Knowing the adversary's strategy, the firm then chooses the prices to maximize its worst-case profit. The advantage of the robust approach is that it does not suffer from the curse of dimensionality inherent in a stochastic dynamic programming approach. Moreover, the optimal value of the robust model is a lower bound on the firm's profit for any demand realization in the uncertainty set.

The demand uncertainty set includes demands of the form $\tilde{D}^t_{mz}(p^t_e, p^t_{bz}) = \tilde{\xi}^t_{mz} \times d^t_{mz}(p^t_e, p^t_{bz})$ for some nonnegative random variable $\tilde{\xi}^t_{mz}$ with mean one. Suppose that $\tilde{\xi}^t_{mz} \in [1-\delta^t_{mz}, 1+\delta^t_{mz}]$ for some $\delta^t_{mz} \le 1$. We can then express any of its realizations as $\xi^t_{mz} = 1 + \delta^t_{mz} w^t_{mz}$ for some $w^t_{mz} \in [-1, 1]$. Hence, w^t_{mz} determines the realization of the random demand $\tilde{D}^t_{mz}$, and it is sufficient to define an uncertainty set for w. If the w^t_{mz} variables can take values independently, the adversary will choose $w^t_{mz} = -1$ for all periods, zones, and channels, resulting in the robust solution setting low prices to reduce the number of unsold units. Thus, we include constraints in the uncertainty set of w:

$$W^t_{\Gamma,\Delta} := \left\{ w \;\middle|\; \begin{array}{ll} -1 \le w^k_{mz} \le 1, & \\ & k = t, \ldots, T,\ \forall z \in [Z], m = e, b, \\ \sum_{m=e,b} \sum_{z \in [Z]} |w^k_{mz}| \le \Gamma^k, & k = 1, \ldots, T \\ \left| \sum_{m=e,b} \sum_{z \in [Z]} a^k_{mz} w^k_{mz} \right| \le \Delta^k, & k = 1, \ldots, T \end{array} \right\}, \tag{8}$$

where $\Gamma, \Delta \ge 0$ are *budgets of uncertainty*, and Γ limits the average absolute percentage deviation of channel-zone demand. Motivated from the common observation that aggregate demand has a lower forecast error than the zone-level demand, Δ limits the aggregate demand percentage deviation. Scalars a^t_{mz} are normalization constants. A choice of small Γ shrinks the uncertainty set; a choice of small Δ permits only a small forecast error in the aggregate demand while allowing larger

zone-level demand errors. To set the parameters for Equation (8), one can set δ^t_{mz} as the absolute percentage error of the channel m zone z demand forecast in previous periods and Δ^t as the error of the chain-level demand forecast. The constants a^t_{mz} can be set as $\delta^t_{mz}\bar{d}^t_{mz}/\sum_{mz}\bar{d}^t_{mz}$ (i.e., demands normalized by δ^t_{mz}), where $\bar{d}^t_{mz}$ are the calibrated demands at observed prices. Given the prices $p=(p^t,\ldots,p^T)$, the adversary chooses demand realizations in $W^t_{\Gamma,\Delta}$ that result in the worst-case profit. The worst-case profit during period t to T of price vector p is

$$U^t_R(p,x^t) := \underset{w\in W^t_{\Gamma,\Delta}}{\text{minimize}} \quad U^t(w,p;x^t), \tag{9}$$

where $U^t(w,p;x^t)$ is the profit during period t to T under realization w given the initial inventory level x^t. Particularly, it is the optimal value of the following linear program:

$$\begin{aligned} &U^t(w,p;x^t)\\ &= \underset{s,y,u}{\text{maximize}} \sum_{k=t}^{T}\sum_{z\in[Z]}(p^k_e s^k_{ez}+p^k_{bz}s^k_{bz})\\ &\qquad -\sum_{z\in[Z]}\left(c_{ez}y_{ez}+\sum_{z'\in[Z]}c_{zz'}y_{zz'}\right)+q\left(u_e+\sum_{z\in[Z]}u_{bz}\right) \end{aligned} \tag{10a}$$

subject to Constraints (6c)–(6f),

$$s^k_{ez}\le d^t_{ez}(p^t_e,p^t_{bz})(1+\delta^t_{ez}w^t_{ez}),\quad k=t,\ldots,T,\ z\in[Z], \tag{10b}$$

$$s^k_{bz}\le d^t_{bz}(p^t_e,p^t_{bz})(1+\delta^t_{bz}w^t_{bz}),\quad k=t,\ldots,T,\ z\in[Z], \tag{10c}$$

where y is the fulfillment vector and s is the sales vector after demand is realized. By LP strong duality, $U^t(w,p;x^t)$ is equivalent to its dual (a minimization LP), and consequently, $U^t_R(p;x^t)$ is the optimal value of a minimization problem with decision variables w and the dual variables of Equation (10). This minimization problem is bilinear, which in general is NP-hard. To overcome this computational challenge, we develop a tractable lower bound on the worst-case profit $U^t_R(p;x^t)$ under the following simple condition on the feasible price set Ω.

Assumption 1. *For any* $p\in\Omega$, $p\ge q+\min\{c_{ez}, \min_{z'\in[Z]}c_{z'z}\}$ *for all* $z\in[Z]$.

Note that if a price p violates the condition of Assumption 1 for some $z\in[Z]$, then the firm will not sell to any online customer from zone z because, regardless of the fulfillment location, it is more profitable to hold on to the inventory and sell it at salvage value. The assumption also implies that $p\ge q$; hence, it is less profitable to hold store inventory than sell to a store customer. Under this assumption, we have the following result.

Proposition 1. *Under Assumption 1,* $U^t_R(p;x^t)\ge U^t_l(p;x^t)$ *for any* $p=(p^t,\ldots,p^T)$, *where*

$$\begin{aligned} &U^t_l(p;x^t)\\ &= \underset{s,y,u,\chi,\upsilon,\psi,\vartheta,f}{\text{maximize}} \sum_{k=t}^{T}\sum_{z\in[Z]}(p^k_e s^k_{ez}+p^k_{bz}s^k_{bz})\\ &\qquad -\sum_{z\in[Z]}\left(c_{ez}y_{ez}+\sum_{z'\in[Z]}c_{z'z}y_{z'z}\right)+q\left(u_e+\sum_{z\in[Z]}u_{bz}\right) \end{aligned} \tag{11a}$$

$$-\sum_{k=t}^{T}(\Gamma^k f^k+\Delta^k|\vartheta^k|) \tag{11b}$$

$$-\sum_{k=t}^{T}\sum_{z\in[Z]}\sum_{m=e,b}(A^k_{mz}|\upsilon^k_{mz}|+|\chi^k_{mz}|)$$

$$\text{subject to}\quad s^k_{mz}\le d^k_{mz}(p)+|\upsilon^k_{mz}|-|\psi^k_{mz}|,\quad m=e,b,\ k=t,\ldots,T,\ \forall z\in[Z] \tag{11c}$$

$$\upsilon^k_{mz}+\psi^k_{mz}=\delta^k_{mz}d^k_{mz}(p^k_e,p^k_{bz}),\quad m=e,b,\ k=t,\ldots,T,\ \forall z\in[Z] \tag{11d}$$

$$|A^k_{mz}\upsilon^k_{mz}-\chi^k_{mz}-a^k_{mz}\vartheta^k|\le f^k,\quad m=e,b,\ k=t,\ldots,T,\ \forall z\in[Z] \tag{11e}$$

Constraints (6c)–(6f),

where $A^k_{bz}:=p^{\max}-q$, $A^k_{ez}:=p^{\max}-q-\min\{c_{ez}, \min_{z'\in[Z]}c_{z'z}\}$, *and* $p^{\max}:=\max\{p:p\in\Omega\}$.

Note that Equation (10) maximizes profit under a given demand realization through the sales and inventory partition variables. On the other hand, Equation (11) maximizes profit with penalty terms through these same variables and auxiliary decision variables. From constraint (11), it is easy to deduce that $|\upsilon^k_{mz}|-|\psi^k_{mz}|$ takes values between the possible demand deviations from $-\delta^k_{mz}d^k_{mz}$ to $+\delta^k_{mz}d^k_{mz}$. Therefore, $U^t_l(p;x^t)$ allows sales, partition variables, and demand realizations to be determined jointly, but a penalty is incurred based on the demand realization.

Algorithm 2. (Pricing Policy R-OCPX)

Require: A vector of initial inventory levels x^1 and feasible price set Ω

Ensure: Determine a sequence of online prices and zone-level store prices

(1) $t\leftarrow 1$
(2) **while** $t\le T$,
(3) Solve (6) for the optimal solution (p^*,s^*,y^*,u^*)
(4) $\hat{p}\leftarrow \arg\max_{p\in\Omega^{t,T}_Z(p^*)}U^t_l(p;x^t)$
(5) $p^t_e\leftarrow\hat{p}^t_e$
(6) $p^t_{bz}\leftarrow\hat{p}^t_{bz}$ for $z\in[Z]$
(7) Exogenously realize demands $D^t(p^t,\xi^t)$, sales s^t, and fulfillments y^t
(8) $x^{t+1}\leftarrow f(x^t,y^t,s^t)$
(9) $t\leftarrow t+1$

We propose Algorithm 2, which we refer to as R-OCPX. In each period, the policy finds prices that maximize the lower bound on the worst-case firm profit $U_l^t(p; x^t)$; that is,

$$V_R^t(x^t) := \max_{p=(p^t,\ldots,p^T)\in\Omega_Z^{t,T}} U_l^t(p; x^t), \tag{12}$$

where $\Omega_Z^{t,T}$ is a feasible price set for $p = (p^t, \ldots, p^T)$. If $\Omega_Z^{t,T}$ includes all feasible prices allowed in Ω, we observe that the resulting pricing policy is too conservative. Hence, we restrict the allowable prices that R-OCPX can choose. In particular, at each period t, R-OCPX first solves Equation (6) for the optimal prices $p^* = (p^{t*}, \ldots, p^{T^*})$. Then $\Omega_Z^{t,T}$ are all the feasible prices that are some neighborhood of p^*. A motivation for this is that in practical large-scale combinatorial optimization instances, such as D-OCPX, there tend to be a pool of (near) alternative optimal solutions. Hence, R-OCPX can preserve the overall D-OCPX expected profit while choosing the maximally robust price vector among these alternatives.

Although the lower bound U_l^t is a nonlinear program, the objective and the constraints (11c) and (11e) can be linearized with standard techniques. On the other hand, V_R^t is a nonlinear program. However, it is possible to reformulate it into a mixed-integer program using the same technique used to derive one.

Lemma 3. *$V_R^t(x^t)$ is equal to the optimal value of a mixed-integer program with $O(I^2)$ binary variables. Under a multinomial logit demand model, the number of binary variables is $O(I)$.*

3.3. Applicability of Policies to Other Omnichannel Retailers

Markdown (or clearance pricing) optimization is a widely used pricing solution to boost profitability while making room for the next season's inventory. In developing the pricing policies, we made several modeling assumptions to tailor the model to the partner retailer's specific business requirements. Although some features are specific to the retailer, the business problem that we described in Section 2 (i.e., inventory is used for multiple channels, but existing pricing systems ignore cross-channel interactions) exists for many major omnichannel retailers. This can lead to reduced sales in one channel and margin erosion in the other, and depending on the retailer, one can be more severe than the other. To offer new solutions for omnichannel retail, IBM Commerce commercialized the omnichannel markdown pricing solution that includes a proprietary implementation of the models in this section. We describe how these models can be adapted to meet the idiosyncratic requirements of other retailers.

The retailer described in this paper allows each store to set its own markdowns but only allows a single e-commerce markdown. In our work with other retailers, we observed different implementations of omnichannel pricing. For example, one may choose to set a single price regardless of the channel or location. These business constraints on prices can be easily modeled through inequalities on prices or on binary variables.

Omnichannel retailers may also differ in their fulfillment methods. Fulfillment has to be accurately modeled by the inventory partition variables, which can be achieved through modifications to the model. For instance, a retailer could have multiple e-fulfillment centers, which can be modeled by adding new supply nodes. Practically motivated constraints on fulfillment, such as SFS capacities per store can also be easily encoded in the model to better approximate fulfillment quantities. Others have a buy online, pick up in store option for their online customers. This can be modeled by assuming this option will be chosen by a known proportion of online customers in a zone, then modifying the constraints (6d) and (6e) appropriately.

4. Pilot Implementation at a Large U.S. Retailer

We next describe the development of an omnichannel price analytics system (part of the IBM Commerce markdown optimization solution). The project proceeded in three phases: (1) a business value assessment on historical data, (2) the development of the omnichannel system, and (3) a commercial release and pilot implementation for clearance pricing at a major U.S. retailer.

4.1. Business Value Assessment

We engaged with the retailer for six months to conduct the business value assessment (BVA) to understand the impact of D-OCPX on representative product categories based on historical data. The first few weeks were spent on working with all stakeholders to define the business problem, select the categories to be analyzed, and collect and process the historical data. Thereafter, we performed the demand model calibration and the value assessment.

The retailer provided transaction log data and inventory data for 195 clearance SKUs in three product categories (notebooks, tablets, and tablet accessories) between January 1, 2014 and December 31, 2014 (see Table 1 for the data summary). We were additionally provided with price information for 18 online competitors. These competitor-price data overlapped with 59 SKUs in our study, and among those that had an overlap, on average, there were six competitors per SKU. We use these data to calibrate the zone-level demand models of each SKU using a procedure we describe in Section EC.2 of the e-companion.

Using the estimated zone-level demand models of each SKU, we optimize the 10- to 12-week clearance

prices through the MIP formulation of Equation (6) with $t = 1$. The retailer's fulfillment strategy was to use the EFC to fulfill the online orders if it had inventory; otherwise, the nearest in-stock store was used unless there were other operational costs and business rules. Fulfillment cost data are not available; hence, we set distance-based shipping costs with additional cost for SFS fulfillment to adhere to priority ordering. For each SKU, the retailer provided the salvage values. We included in Equation (6) pricing constraints based on SKU-specific rules provided by the retailer, such as a minimum time between markdowns, minimum and maximum markdown price percentages, and price bounds. Model (6) was developed as a Java API, which was evaluated on an OS X computer with an Intel Core i7 processor. CPLEX 12.6.2 with its default termination criteria was used to solve the MIPs. Model (6) was solved with up to 20 price discretizations. Problem instances had up to 10K binary variables, 50K flow variables, and 100K constraints. In comparison, the prepilot approach was to solve price optimization models for each channel and store(s) separately, which are significantly smaller in size. Figure EC.1 in the e-companion provides the distribution of the computational runtimes of the MIPs, the percentage gap of the MIP optimal solution to the LP root node, and the total number of branch-and-bound nodes in solving the MIP problem. We note that 80% of the instances solved in less than 40 seconds with a gap of at most 1% with the root node LP with little or no branching required. Thus, our MIP-based approach allows us to solve the large-scale problem within the required business cycle.

Because the "true" demand values and fulfillment quantities are unknown for the counterfactual prices, we use the revenue component of V_D^1 in Equation (6) as the predicted clearance revenue of D-OCPX. Table 2 shows the predictions on D-OCPX average channel prices (normalized by regular price), category-wide sales, and category-wide clearance revenues. We make the following observations. First, D-OCPX results in lower online prices than the historical online prices, resulting in a projected increase in online sales from customers buying online instead of from the store. Higher online sales is desirable because, unlike store sales that can only be fulfilled through same-zone stores, online sales can be directed by OMS to any store with slow-moving inventory. A benefit of D-OCPX is that it uses prices to manage channel demand in accordance with partitions that rebalance inventory in the network. In stores with slow-moving inventory, SFS has a large partition, resulting in a low store markdown. In stores with strong sales, the store markdown is also low to maximize profitability from store sales. Thus, our second observation: D-OCPX results in higher average store prices than the historical average store prices. Hence, D-OCPX is addressing the concern of margin erosion because of the low store prices recommended by the legacy MDO system. In fact, our demand models predict more than 21% reduction in lost store-sales opportunities at the recommended prices across categories. The net effect of the D-OCPX prices and the predicted fulfillment yields an increase in clearance revenues ranging between 6% and 12%, a combined increase of \$12.5 million in the total clearance revenue of the 195 SKUs.

The BVA was presented to the retailer and reviewed by their team of pricing analysts and senior executives who were well versed with the incumbent MDO system. Their feedback was overwhelmingly positive. The pricing analysts have long recognized that the incumbent system fails for SKUs with significant cross-channel (SFS) fulfillment because of its inability to partition inventory between store demand and online fulfillment, and hence, they requested to productize the OCPX solutions immediately.

4.2. Development of a Commercial Solution

A proprietary version of an OCPX solution was approved for commercial deployment at IBM. To productize the OCPX solution, we collaborated with the IBM Commerce team who developed the necessary IT infrastructure, data processing, analytics integration, and user interface.

The team conducted system integration testing of the OCPX system to validate that the system performs in accordance with user expectations. This was done by comparing, on the same SKU, the price outputs of the OCPX system and of the incumbent MDO system. There were 43 clearance SKUs (across 10 consumer electronics categories) during 2015 Q4 through 2016 Q1 selected for the system integration test. Unlike the tests conducted with historical clearance data, these SKUs were soon to enter their clearance period. The test revealed that, compared with the incumbent system,

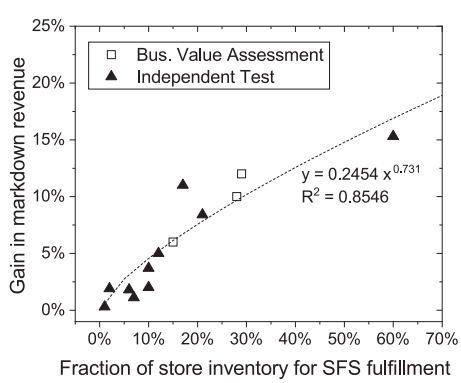
Table 2. Impact of D-OCPX on Average Channel Prices and Category-wide Sales in Business Value Assessment

	Store price		Online price		Percentage change in sales			
Category	Actual, %	D-OCPX, %	Actual, %	D-OCPX, %	Online	Store	Total	Change in clearance revenue, %
Notebooks	81	93	90	85	+16	–1	+3	+10 (+\$9.0 million)
Tablets	81	90	79	77	+15	+10	+11	+12 (+\$2.3 million)
Accessories	81	93	90	85	+11	+5	+6	+6 (+\$1.2 million)

the OCPX system lowered store markdowns by two percentage points on average without any drop in the system predicted sell-through rate. The lower OCPX store markdowns, an important metric addressing the retailer's main concern of margin erosion, were consistent with the observations from the BVA. For each SKU, we also constructed a vector whose elements are the maximum markdowns of each store. We observed that the price dispersion in stores, which is empirically measured as the coefficient of variation of this vector, was up to 6.45% lower in the OCPX prices compared with incumbent MDO prices. This suggests OCPX results in a higher degree of price parity in the retail network.

As part of the system-integration test, projected revenues resulting from the price outputs of OCPX and the incumbent MDO were also computed. The retailer had years of experience with the incumbent demand forecasting engine and its accuracy for various categories. Hence, for unbiased revenue projections, it was determined that the projected revenue of both price outputs should be estimated from the incumbent MDO system with inventory adjusted (using the OCPX partitions) to account for cross-channel fulfillment in OCPX. No adjustments were made to compute the revenue from the incumbent prices because its undiscounted online price results in low online sales and, hence, low fulfillment. The difference between the two revenue projections provides a projection of the revenue gain from the OCPX system. We plot the projected gain against the predicted degree of supply-side channel interdependency (Figure 3) and observe that *the revenue gain is expected to be higher if cross-channel fulfillment is used more*. The horizontal axis of the figure is the OCPX partition for SFS (aggregated over SKUs in a category) divided by the aggregate store inventory, which is the predicted proportion of store inventory used for online fulfillment. The vertical axis is category-wide revenue gain from OCPX, which is computed as the relative difference between the OCPX projected revenue (aggregated over the category) and the incumbent MDO projected revenue (aggregated over the category).

Figure 3. Gain in Markdown Revenue Plotted Against Store Inventory Partitions for SFS Fulfillment



Note. Each data point represents a product category in the system-integration test.

4.3. Commercial Release and Pilot Implementation

After a successful system-integration test, a limited commercial release of the proprietary version of the OCPX system went live in March 2016 with the partner retailer as its pilot client. This was followed with its general availability in May 2016 with the release first announced at the IBM Amplify 2016 conference.

Subsequent to the commercial release in Q2 2016, the retailer has gradually transitioned SKUs marked for clearance to the OCPX system. For the transitioned SKUs, the parameters of the omnichannel demand model are periodically recalibrated, and the OCPX models are reoptimized every period. The SKUs were monitored in real time by IBM Commerce as part of continued performance testing. For these SKUs, the optimality of the recommended prices depends on how accurately the OCPX inventory partitions predict the actual store inventory quantities used for SFS fulfillment. Hence, an inventory partition accuracy metric is additionally reported to the retailer. This accuracy metric compares the OCPX's week 1 inventory partitions against the actual fulfillments during the same weeks. For 22 SKUs across five consumer electronics categories, the mean absolute percentage error at the chain level was about 20%. Closing this gap requires the modeling of the market-basket fulfillment and complex operational factors and is a topic for future research.

4.3.1. Causal Model Analysis. OCPX optimizes across the complete retail network and uses as input the inventory levels across the 1,000+ stores. Hence, it is impractical to estimate the treatment effect of OCPX through finding close substitutes (similar to Caro and Gallien 2012) because not only should the identified substitutes have similar demand characteristics, but the distribution of inventory at the start of clearance should also be identical. Furthermore, a SKU is only in clearance for one season before being retired. Hence, we cannot apply pretreatment (legacy) and posttreatment (OCPX) on the same SKU in different clearance seasons. More crucially, we could not provide input into the selection of SKUs that were transitioned to the new OCPX system.

Hence, to estimate the revenue improvement of OCPX from the pilot implementation, our approach was to analyze the pilot data through causal model analysis via statistical adjustment using appropriate pretreatment predictors (Gelman and Hill 2006, chapter 9). This is a variation of the difference-in-differences test (see Fisher et al. 2017), wherein we

evaluate the change in the markdown season revenue rate between the treatment SKUs and the control SKUs by using the regular season revenue rate as a baseline for both and adjusting for other pretreatment factors.

We obtained one-year historical data from the IBM Commerce production system for a sample of SKUs across 34 categories with which each SKU either had the OCPX treatment or not. All SKUs were marked for clearance in both channels with a markdown end date (last date sold) in Q1 of 2017. We selected SKUs for which the full clearance season is in the data. For an average treatment SKU, the online sales share is 12.6% during the regular season, and this nearly doubles to 24.1% during clearance season. In addition, nearly 98% of online sales during clearance season is fulfilled with SFS using approximately 20% of store inventory.

To identify the presence of selection bias in the SKUs in control and treatment data, we compute a variety of regular season (pretreatment) covariates, such as the ticket price, average selling price, sales rate, revenue rate, and channel shares for all SKUs. We observe that the range of the respective distributions across the control and treatment SKUs has a significant overlap. To formalize, we use a nonparametric preprocessing matching method called MatchIt (Ho et al. 2007, 2011) that estimates a propensity score—defined as the probability of receiving the treatment given the pretreatment covariates—and matches observations in treatment with those in control by balancing the propensity score (and, thus, the covariate distribution) to reduce the selection bias in parametric causal inference. We observe that higher propensity products correspond to those with high regular season promotion depths, strong sales, and a large online presence. We expect OCPX to yield relatively higher benefits for higher propensity-score items and lower benefits for lower propensity-score items characterized by weaker sales rates and a small online presence. This indicates that there is a selection bias, but the left-side panels of Figure 4 that plot the raw density of the propensity scores for the treated and control SKUs reveal that this bias is not severe. The average propensity score is 0.2 for control and is slightly higher at 0.23 for the treatment SKUs. Next, we use MatchIt to select control SKUs in the midrange scores (between 0.15 and 0.35) with the nearest propensity score to the treatment SKUs and discard the remaining unmatched SKUs. The distributions of the propensity scores after matching are plotted in the right-side panels of Figure 4, showing nearly identical distributions with an average propensity score of 0.24 each. For this matched subclass, we conclude that there is no selection bias based on the descriptive covariates of the SKUs. To confirm this, recomputing propensity scores on this matched subclass results in scores of 0.5 ± 0.02 with 92% confidence.

We use the following regression model to estimate the average causal effect of OCPX (across all SKUs and within the matched subclass); each SKU is an observation:

$$\begin{aligned}&\ln(\text{Avg-Weekly-MD-Rev}_i)\\&\quad \sim \alpha_0 + \alpha_1 \ln(\text{Avg-Weekly-Reg-Rev}_i)\\&\quad + \alpha_2\,\text{Avg-Weekly-MD-Inventory}_i\\&\quad + \alpha_3\,\text{Reg-Online-Share}_i + \alpha_4\,\text{Treatment}_i\\&\quad + \alpha_5\,\text{Treatment}_i * \text{Reg-Online-Share}_i + \epsilon \quad \forall i \in \text{Items}.\end{aligned} \tag{13}$$

The terms Avg, MD, Reg, and Rev correspond to average, markdown, regular, and revenue, respectively. Here, $\text{Treatment}_i = 1$ if the SKU i used OCPX and zero otherwise. The average weekly markdown and regular season revenues are obtained by dividing the total revenue in each season by their respective durations, and the average weekly markdown inventory is the initial clearance inventory divided by markdown duration. This normalization was done because the durations can vary by season for the same item and across items.

Figure 4. (Color online) Histograms of the Density of Propensity Scores for Observations Before and After Matching

The dependent variable in the regression model is the log markdown revenue rate. Motivated by the observation in Figure 3 that the gain in OCPX markdown revenue is proportional to the use of SFS fulfillment, we interact the regular season online share with the treatment effect (for this retailer, the regular season online share is highly correlated with the SFS fulfillment during clearance). We also add the treatment variable by itself to gather local effects. Moreover, for both treatment and control SKUs, we expect that the higher the online presence, the retailer can clear the inventory better using SFS fulfillment. We also control for pretreatment predictors, which are the log regular season revenue rate and the normalized initial clearance inventory.

The coefficients of interest are α_4 and α_5. Table 3 shows the estimation results for the data set with all SKUs and for the data set with only the matched SKUs. For each data set, we present the results of three regression models: with the main treatment effect only, with the interaction treatment effect only, and with both. For the matched SKUs, the coefficient for Reg-Online-Share (α_3) was not significant in all the three models; hence, we report a model without it. The table also reports results from an F-test for joint significance of α_4 and α_5 in which the null hypothesis is that both α_4 and α_5 are zero. Based on the p-values of the F-test, we can reject the null hypothesis on the raw data and the matched data. We observe that the coefficient estimates of the treatment effect (α_4, α_5) has the same sign across models. To compute average improvement because of OCPX on the treatment SKUs, we compute the average of the SKU-level benefits over all treatment SKUs. We note that among the three models, the one with only the interaction term results in the lowest gains, which is what we report. It is 13.7% for the data set with all SKUs and 20%–24% (the former when we include the Reg-Online-Share control variable) for the model with matched SKUs only. We observe higher gains for the treatment SKUs in the matched subclass because the average sales and average online share (kept the same for control SKUs in the matched subclass) is slightly higher compared with the treatment SKUs in the raw data.

5. Simulation Experiments

The previous section describes a pilot implementation of the omnichannel pricing system at a U.S. retailer, resulting in an estimated revenue increase compared with the legacy system. We next conduct a comparison of the omnichannel policies and the legacy policy under a simulation environment.

5.1. Parameter Generation

We randomly generate 27 problem instances in which each instance has randomly chosen parameters for the demand models and the zone-to-zone distances. We set $Z = 5$ and $T = 5$. To model the frequently observed case that EFC is depleted at the start of a product's clearance period, we set $x_e = 0$. To model high-ticket items sold by the partner retailer, we set the minimum and maximum price for the product as \$125 and \$500, respectively. We discretize the set of feasible prices into a finite set of 31 prices with a stepsize of \$12.50. The salvage value for unsold items is \$50. The zone-to-zone distance matrix is populated with random distances from zero to 2,500 miles. We then set the shipping cost between zones as $\$4.67 + \$0.001 \times d$, where d is the interzone distance in miles. Note that the shipping costs can take values between \$4.67 and \$7.17.

Table 3. Causal Model Estimated from the Pilot Data

		All SKUs			Matched SKUs		
ln(Avg-Weekly-MD-Rev)	Coefficient	(1)	(2)	(3)	(1)	(2)	(3)
Constant	α_0	1.820***	1.776***	1.819***	1.202**	1.078**	1.191**
		(0.178)	(0.169)	(0.178)	(0.374)	(0.364)	(0.373)
ln(Avg-Weekly-Reg-Rev)	α_1	0.597***	0.598***	0.594***	0.664***	0.672***	0.657***
		(0.021)	(0.021)	(0.021)	(0.038)	(0.037)	(0.039)
Avg-Weekly-MD-Inventory	α_2	0.004***	0.004***	0.004***	0.004***	0.004***	0.004***
		(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Reg-Online-Share	α_3	0.563**	0.734**	0.636**			
		(0.263)	(0.240)	(0.270)			
Treatment	α_4		0.195**	0.140		0.325**	0.221
			(0.097)	(0.118)		(0.158)	(0.178)
Treatment × Reg-Online-Share	α_5	0.918*		0.497	1.743**		0.119
		(0.509)		(0.619)	(0.843)		(0.950)
Number of observations		275	275	275	78	78	78
Number of treatment		57	57	57	39	39	39
R^2		0.832	0.832	0.833	0.850	0.850	0.853
F-test for joint significance of α_4 and α_5		*	**	*	**	**	*

Notes. SKUs with very short clearance periods and rate of sales were eliminated. Standard errors are in parentheses.
***$p < .001$; **$p < .05$; *$p < 0.1$.

Given the chosen zone z channel prices at time t, the stochastic demand in channel m is

$$\tilde{D}^t_{mz}(p_e, p_b) = N^t_z \times \theta_{mz}(p_e, p_b) \times (1 + \tilde{w}_{mz}\delta_m), \quad \text{for } m = e, b, \tag{14}$$

where θ^t_{mz} is the deterministic MNL function (7). The demand stochasticity originates from the random demand factor $\tilde{\xi}^t_{mz} = 1 + \tilde{w}_{mz}\delta_m$ for all t, where δ_e, δ_b are parameters and $\tilde{w}_{mz}$ is a random variable with mean of zero. We chose $\delta_b = 0.5$ and $\delta_e = 0.9$ to reflect our observations from the retailer's data that the forecast error is higher for online demand. Parameter α_{mz} in Equation (7) is randomly chosen from the interval [10.8, 13.2]. The online price coefficient β_{ez} in Equation (7) is randomly chosen from [0.0243, 0.0297], and the store price coefficient β_{bz} is randomly chosen from [0.0225, 0.0275]. Note that if both channel prices are \$350, then the average own and cross-elasticities of the online channel are –6.1 and 5.3, respectively; for the store channel, they are –3.4 and 3.3, respectively. These high elasticity values are typical of products marked for clearance. We set

$$N^t_z = B\left(\frac{t}{T+1}; a, b\right) \times \varepsilon_z \times \sigma, \quad t \in [T], z \in [Z], \tag{15}$$

where B is the pdf of a beta random variable with randomly chosen shape parameters $b \in [1, 5]$ and $a \in [1, b]$, ε_z is randomly chosen in [0,1], and σ is a scaling factor that ensures $\sum_z \sum_t N^t_z = 100$. Thus, in all problem instances, the aggregate market size is always 100. To complete the specification of a problem instance, we set $x_{bz} = \sum_t N^t_z$; that is, the available inventory at zone z is set to the expected market size at z. Note that the aggregate demand in zone z can be greater than $\sum_t N^t_z$ under some demand realizations.

The zero-mean random variable $\tilde{w}_{mz}$ has the distribution illustrated in Figure EC.2 and can take values between –1 and 1. We ensure the realizations of w have the property

$$\sum_{m=e,b} \sum_z \dot{w}_{mz} = 0. \tag{16}$$

This models our observations that although the zone-level channel demand may deviate significantly from its expected value, the expected aggregate market size is close to its actual value. Given the prices, the set of demand factors $(w_{mz})_{mz}$ determines the actual demand realizations. To conduct the simulation experiments, we randomly generate 10^3 sets of the demand factors.

5.2. Methodology

For any pricing policy, our simulations construct the distribution of realized profits (total revenue minus fulfillment costs) when the policy is applied to a specific problem instance. The distribution is constructed from the 10^3 samples of the demand factors $(w_{mz})_{mz}$.

We next describe how we compute the realized profit under a given sample of demand factors. For each week t starting from week 1, the policy sets channel prices p^t_e, $(p^t_{bz})_{z\in Z}$ based on the current inventory levels. The actual demand $(D^t_{ez}, D^t_{bz})_{z\in Z}$ is realized based on Equation (14). The channel sales, fulfillment decisions, and profit realization are then determined by a fulfillment engine, which returns the inventory levels for week $t+1$. The fulfillment engine takes in as input the week t realized demand, the week t prices, and the week t store inventory levels. The engine was designed to approximate how fulfilment is done at the partner retailer: store customers arrive throughout the day, and fulfillment of online orders is determined periodically (e.g., after one day). If week t is divided into K fulfillment periods (e.g., seven days), we assume that the realized week t zone-level demand is uniformly distributed over the K periods. At the end of the kth period, the zone-level store demands are met to the maximum extent possible using the same-zone store inventory. The remaining inventories in the Z stores are then used to fulfill the kth period online demands through an optimization model that determines online sales and fulfillment with the objective of maximizing the myopic revenue minus fulfillment costs. The total profit is the sum of each week's profit and the salvage value from unsold items at the end of week 5.

We tested policies D-OCPX and R-OCPX using the methodology described. The parameters of the uncertainty set (8) used are $\frac{\Gamma^k}{2Z} = 0.1$, $a^t_{mz} = \delta^t_{mz}$, and $\Delta^t = 0$. R-OCPX constructs its feasible price set $\Omega^{t,T}_Z$ for Equation (12) by setting lower and upper bounds that are 95% and 105%, respectively, of the optimal price solution to Equation (6). We additionally test the following pricing policies based on the legacy-pricing model:

- Policy legacy sets the online price and store prices at period t as

$$p^t_e \leftarrow \arg\max_{p\in\Omega} p \cdot \min\left(x^t_e, \sum_{z\in[Z]} N^t_z \cdot \theta_{ez}(p, p^{nom})\right) \tag{17}$$

$$p^t_{bz} \leftarrow \arg\max_{p\in\Omega} p \cdot \min(x^t_{bz}, N^t_z \cdot \theta_{bz}(p^{nom}, p)), \quad z \in [Z], \tag{18}$$

where $p^{nom} = \$350$, a price level that is approximately the center of the feasible price range. The policy ignores any cross-channel interactions when optimizing the channel prices. The effect of the online price on stores is ignored, and conversely, the effect of store prices on the online demand is ignored. The full store inventory is used to determine store prices. This mimics the output of the legacy system. Because $x^t_e = 0$, p^t_e is set to \$500 (no markdown).

- Policy legacy + adjust sets the store prices in period t according to Equation (18). Then, to set the online price, it solves Equation (17) but with the adjustment $x_e^t \leftarrow x_e^t + \nu \sum_{z\in[Z]} x_{bz}^t$ for some parameter $\nu \geq 0$. This policy mimics the manual adjustment process of the merchandise managers to the inputs of the legacy-pricing system. Because $x_e^t = 0$, we chose $\nu = 0.27$, which is the average MNL online share if both channel prices are \$350. The store price is unadjusted because, based on our conversations, managers accept the store markdowns from the legacy system as correct but take action on the undiscounted online price.
- Policy legacy + OCPX sets the period t online and store prices based on Equations (17) and (18) but with the adjustments $x_e^t \leftarrow x_e^t + \sum_{z,z'\in[Z]} y_{zz'}^*$, and $x_{bz}^t \leftarrow x_{bz}^t - \sum_{z'\in[Z]} y_{zz'}^*$, where y^* is the optimal inventory partition from Equation (6). This policy model ignores cross-channel demand interactions but partitions store inventory to account for cross-channel fulfillment.

We implemented the computational experiments in Python 3.5. The optimization models are solved using CPLEX Optimization Studio 12.7. We set CPLEX to terminate when an integer feasible solution is within 0.01% of optimality. The average runtime of D-OCPX and R-OCPX on a sample path are 6.796 and 7.895 seconds, respectively. The average run time of legacy, legacy + adjust, and legacy + OCPX are 1.025, 1.712, and 11.046 seconds, respectively.

5.3. Results

The expected profit of the optimal pricing policy cannot be computed and is also difficult to approximate using sample average approximation because the number of variables and constraints of the optimization model is proportional to the sample size. Moreover, because the stochastic channel demand does not follow a multinomial distribution, the upper bound in Lemma 1 does not apply. Nevertheless, the following lemma gives a simple and easily computable upper bound on the optimal profit, which we use to benchmark the pricing policies.

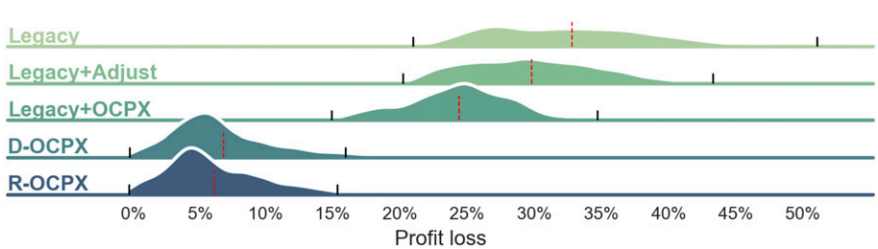
Lemma 4. *$V^1(x) \leq V_{PF} := E_{\tilde{w}}[U_{PF}(\tilde{w})]$, where $U_{PF}(w) := \max_{p=(p^1,\ldots,p^T)} U^1(w,p;x)$, and U^1 is the optimal value of optimization model (10) with $t = 1$.*

Figure 5 shows the distributions of profit loss in one specific problem instance (the distributions in all 27 instances follow a similar pattern). The profit loss on a sample path w is defined as the relative difference between the realized profit of a pricing policy and $U_{PF}(w)$. We observe that the pricing policies based on the legacy model have a significantly higher profit loss compared with the omnichannel pricing policies. This is further demonstrated in Figure 6, which uses scatterplots to compare the *average* profit loss of the legacy-pricing policies to that of D-OCPX in all 27 problem instances. The average profit loss of D-OCPX is between 5% and 8%. The prices from the unadjusted legacy model (legacy) results in roughly 20%–40% average profit loss. Adjusting the pricing policy by injecting online inventory (legacy + adjust) reduces the average profit loss slightly. Out of the legacy models, the one that incurs the lowest average profit loss is legacy + OCPX, which has average profit losses of 15%–30%.

To understand the reason behind these results, we next compare and contrast the pricing policies with respect to their average markdowns and their average sales. Table 4 presents these statistics for the problem instance corresponding to Figure 5. From the table, we observe that policies legacy and legacy + adjust result in only a small percentage of inventory sold through the online channel. This is because of the significantly larger store markdown compared with the online markdown. Because customers consider the online channel and store channel as substitutes, this price difference induces a larger proportion of customers to purchase from stores. Because of the large store markdowns, the resulting effect is a low overall profit margin.

The legacy store markdown-optimization model assumes that the store inventory input will be sold only through the store channel. Hence, the large store markdowns of legacy and legacy + adjust are the result of the unadjusted store inventory inputs to the legacy model. But based on Table 4, it is optimal to sell an average of 42% of the store inventory through the online channel. Because the D-OCPX model (6) optimizes the inventory partitions that allocate store inventory between the channels, we also tested the legacy + OCPX

Figure 5. (Color online) The Profit Loss Distribution Under the Pricing Policies in One Problem Instance



Notes. The kernel density is estimated from the profit loss in 10^3 sample paths. The black vertical lines mark the range (removing statistical outliers), and the dashed vertical lines (red online) mark the mean of the distribution.

Figure 6. (Color online) Scatterplot of D-OCPX Average Profit Loss and a Baseline Policy (Legacy, Legacy + Adjust, Legacy + OCPX) Average Profit Loss

Note. Points to the right of the dashed line (red online) are problem instances in which D-OCPX has a lower average profit loss compared with the baseline.

pricing policy, which adjusts the inventory inputs to the legacy model based on the optimized partitions from Equation (6). With this adjustment, the average online and store markdowns are brought closer together (see Table 4), implying that the inventory partitions of Equation (6) correctly identify a balanced allocation of inventory between channels. However, the average channel markdowns of 35% and 39% with legacy + OCPX are larger than the perfect foresight markdowns. Thus, even if inventory is balanced across the network, a markdown-optimization model that ignores channel substitution and does not jointly optimize omnichannel prices results in significant profit loss.

Figure 7 shows the distribution (over the 10^3 samples) of profit increase resulting from R-OCPX with a box plot for each problem instance. The profit increase was computed as the D-OCPX profit loss minus the R-OCPX profit loss. The dashed line in the figure denotes no improvement in profit. In the first nine problem instances shown in the figure, R-OCPX has a higher profit than D-OCPX in about 75% of the samples. In almost all instances, the median R-OCPX profit increase is positive. We investigate which cases result in the largest profit increase by R-OCPX. Define the online demand factor $\bar{\xi}_e := 1 + \delta_e \bar{w}_e$, where $\bar{w}_e = \frac{1}{Z}\sum_{z\in[Z]} w_{ez}$ is the average online percentage deviation. Figure 8 shows scatterplots of the R-OCPX profit increase plotted against $\bar{\xi}_e$. Note that $\bar{\xi}_e < 1$ if the online demand is lower than its expected value. Because of the procedure for generating the demand factor samples in Equation (16), the average store percentage deviation is $-\bar{w}_e$. In other words, if the average online demand is higher than expected (i.e., $\bar{\xi}_e > 1$), then the average store demand is lower than expected. It is clear from these scatterplots that the largest profit increase resulting from R-OCPX occurs whenever the average online factor is small, that is, when the actual online demand is lower than the predicted values and actual store demand is higher than predicted.

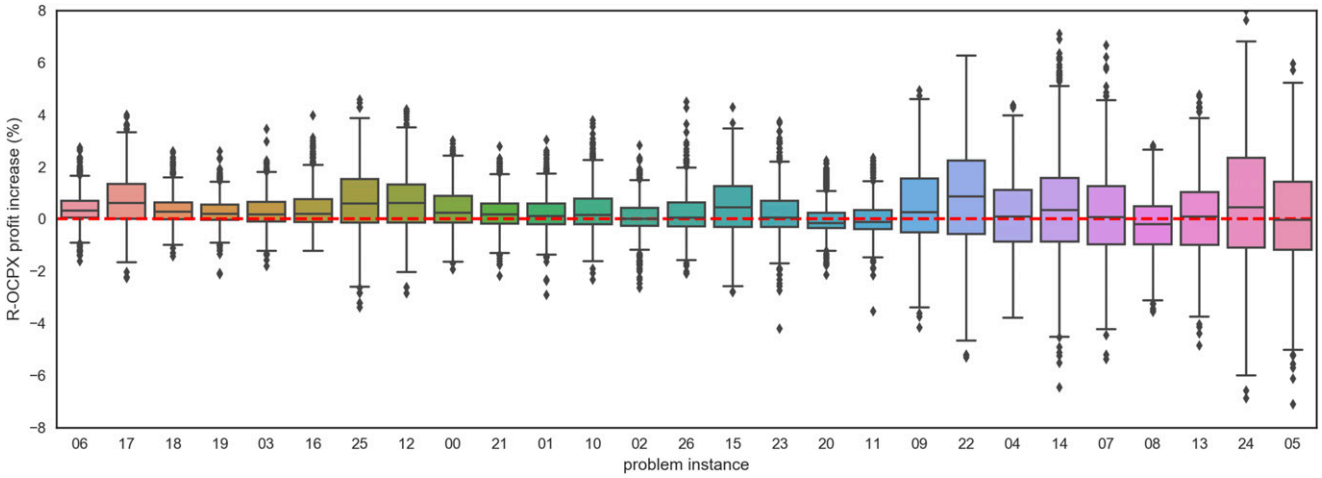
Observe that the omnichannel demand uncertainty has an asymmetric impact on a retailer's expected profitability. The reason is, when store demand is lower than predicted and online demand is higher than forecasted, the omnichannel retailer can gainfully employ SFS fulfillment to satisfy the excess online demand using unused store inventory, resulting in a win–win revenue situation. On the other hand, when store sales are higher than expected (exceeding store inventory), and online demand is correspondingly lower, this results in lost in-store sales, and the SFS

Table 4. The Average Price Markdown, Average Sales, and Average Unsold Units of Different Pricing Policies Applied to a Specific Problem Instance

	Average price markdown		Average inventory usage		
	Online, %	Store, %	Online sales, %	Store sales, %	Unsold, %
Legacy	0	41	0.5	85	14
Legacy + adjust	26	41	10	79	11
Legacy + OCPX	35	39	32	59	10
D-OCPX	16	12	40	48	12
R-OCPX	13	11	36	50	14
Perfect foresight	11	10	42	54	4

Note. The average is estimated from 1,000 sample paths.

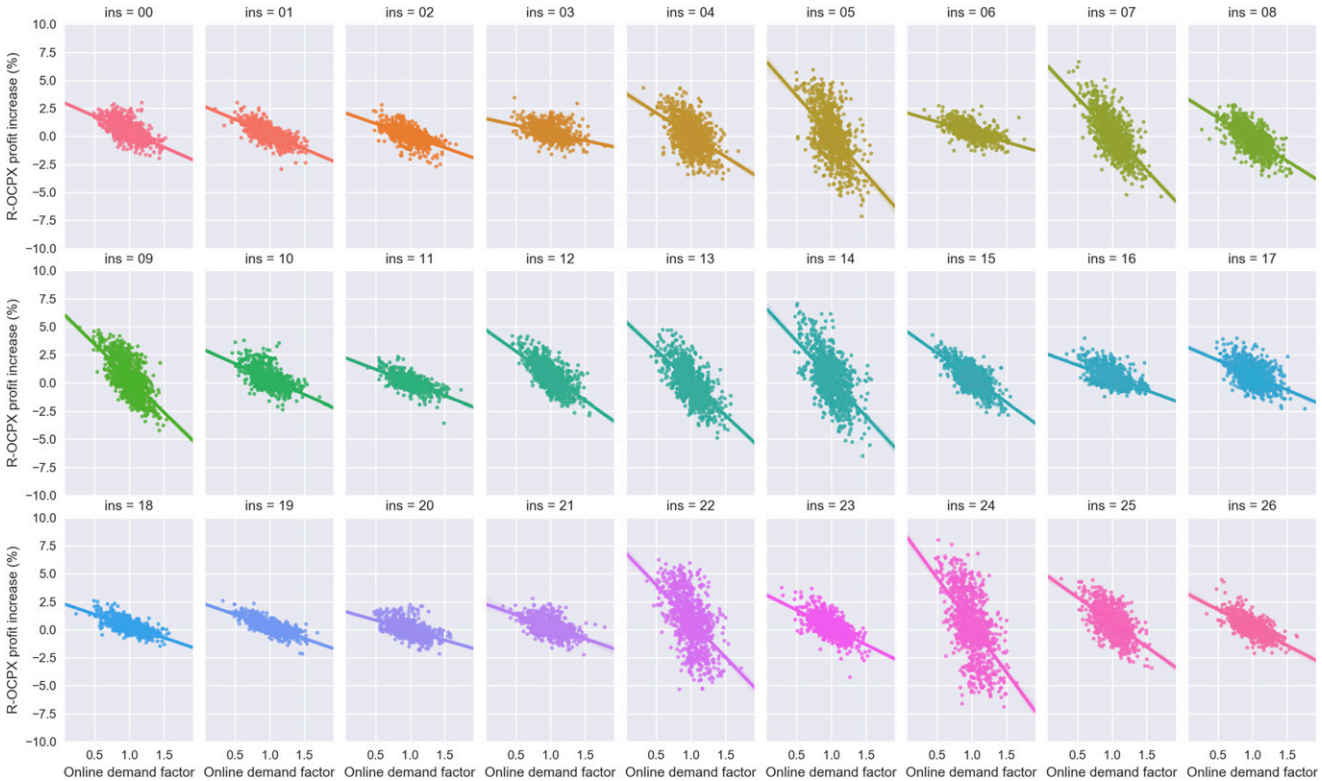
Figure 7. (Color online) Boxplots of the R-OCPX Profit Increase in 10^3 Samples



Note. The boxplots have been ordered in decreasing 25% quantile.

option is also of limited use because of weak online demand, resulting in a loss–loss in both channels. Therefore, the robust adversary can be expected to choose low online demand and high store demand (particularly, higher store demand for stores with low inventory). As can be seen in Figure 8, the proposed R-OCPX solution increasingly outperforms D-OCPX when omnichannel demand realizations are highly adversarial ($\bar{\xi}_e$ is very small). This also explains why R-OCPX profitably protects against worst-case demand realization by optimally raising prices relative to D-OCPX (see Table 4).

Figure 8. (Color online) Scatterplots and Linear Fit of the Profit Increase by R-OCPX Against the Average Online Demand Factor



Note. The plots demonstrate a decreasing relationship between the profit increase and the average online factor.

Acknowledgments

The authors thank the omnichannel retailer who wishes to remain anonymous and, more specifically, its pricing team, who provided the key business motivation and insights that helped in shaping the solution design. The commercialization of this work is a result of the combined efforts that includes the following team members from the IBM Commerce business unit: Saibal Bhattacharya, Irina Fedulova, Vitaly Grechko, Nikolay Murzin, Manjunath Pandit, Emily Port, Erich Schellhas, Oleg Sidorkin, Mahesh Virupakshaiah, Maloney Whitfield, Jin Jing Xie, Emrah Zarifoglu, and Pavel Zelinsky. This list includes the data scientists, product managers, and client executives. The authors also thank Markus Ettl and Suzanne Valentine who provided executive support through the project. The authors also thank Professor Yan Huang for valuable suggestions related to propensity score analysis for the causal model.

References

Acimovic J, Graves SC (2014) Making better fulfillment decisions on the fly in an online retail environment. *Manufacturing Service Oper. Management* 17(1):34–51.

Adida E, Perakis G (2006) A robust optimization approach to dynamic pricing and inventory control with no backorders. *Math. Programming* 107(1):97–129.

Bitran G, Caldentey R (2003) An overview of pricing models for revenue management. *Manufacturing Service Oper. Management* 5(3):203–229.

Bitran G, Mondschein SV (1997) Periodic pricing of seasonal products in retailing. *Management Sci.* 43(1):64–79.

Bitran G, Caldentey R, Mondschein S (1998) Coordinating clearance markdown sales of seasonal products in retail chains. *Oper. Res.* 46(5):609–624.

Bront JJM, Méndez-Díaz I, Vulcano G (2009) A column generation algorithm for choice-based network revenue management. *Oper. Res.* 57(3):769–784.

Caro F, Gallien J (2012) Clearance pricing optimization for a fast-fashion retailer. *Oper. Res.* 60(6):1404–1422.

Chen X, Simchi-Levi D (2012) Pricing and inventory management. Philips P, Özer Ö, eds. *Oxford Handbook of Pricing Management* (Oxford University Press, Oxford, UK), 784–822.

Elmaghraby W, Keskinocak P (2003) Dynamic pricing in the presence of inventory considerations: Research overview, current practices, and future directions. *Management Sci.* 49(10):1287–1309.

Fisher M, Gallino S, Li J (2017) Competition-based dynamic pricing in online retailing: A methodology validated with field experiments. *Management Sci.* 64(6):2496–2514.

Gallego G, Phillips R (2004) Revenue management of flexible products. *Manufacturing Service Oper. Management* 6(4):321–337.

Gallego G, van Ryzin G (1994) Optimal dynamic pricing of inventories with stochastic demand over finite horizons. *Management Sci.* 40(8):999–1020.

Gallego G, van Ryzin G (1997) A multiproduct dynamic pricing problem and its applications to network yield management. *Oper. Res.* 45(1):24–41.

Gelman A, Hill J (2006) *Data Analysis Using Regression and Multilevel/Hierarchical Models* (Cambridge University Press, New York).

Grewal D, Ailawadi KL, Gauri D, Hall K, Kopalle P, Robertson JR (2011) Innovations in retail pricing and promotions. *J. Retailing* 87(Suppl. 1):S43–S52.

Harsha P, Subramanian S, Ettl M (2019) A practical price optimization approach for omni-channel retailing. *J. Optim.* Forthcoming.

Ho DE, Imai K, King G, Stuart EA (2007) Matching as nonparametric preprocessing for reducing model dependence in parametric causal inference. *Political Anal.* 15(3):199–236.

Ho DE, Imai K, King G, Stuart EA (2011) Matchit: Nonparametric preprocessing for parametric causal inference. *J. Statist. Software* 42(8):1–28.

Jasin S, Kumar S (2013) Analysis of deterministic LP-based booking limit and bid price controls for revenue management. *Oper. Res.* 61(6):1312–1320.

Jasin S, Sinha A (2015) An LP-based correlated rounding scheme for multi-item ecommerce order fulfillment. *Oper. Res.* 63(6):1336–1351.

Lan Y, Gao H, Ball MO, Karaesmen I (2008) Revenue management with limited demand information. *Management Sci.* 54(9):1594–1609.

Lei Y, Jasin S, Sinha A (2018) Dynamic joint pricing and order fulfillment for e-commerce retailers. *Manufacturing Service Oper. Management* 20(2):269–284.

Liu Q, van Ryzin G (2008) On the choice-based linear programming model for network revenue management. *Manufacturing Service Oper. Management* 10(2):288–310.

Maglaras C, Meissner J (2006) Dynamic pricing strategies for multiproduct revenue management problems. *Manufacturing Service Oper. Management* 8(2):136–148.

Perakis G, Roels G (2008) Regret in the newsvendor model with partial information. *Oper. Res.* 56(1):188–203.

Talluri K, van Ryzin G (1999) A randomized linear programming method for computing network bid prices. *Transportation Sci.* 33(2):207–216.

Thiele A (2009) Multi-product pricing via robust optimisation. *J. Revenue Pricing Management* 8(1):67–80.

Topaloglu H (2009) Using Lagrangian relaxation to compute capacity-dependent bid prices in network revenue management. *Oper. Res.* 57(3):637–649.

Zhang J, Farris PW, Irvin JW, Kushwaha T, Steenburgh TJ, Weitz BA (2010) Crafting integrated multichannel retailing strategies. *J. Interactive Marketing* 24(2):168–180.

Zhao W, Zheng Y-S (2000) Optimal dynamic pricing for perishable assets with nonhomogeneous demand. *Management Sci.* 46(3):375–388.